

Frozen replay contract $D, \tilde{D}, y_t, g, r$, and τ remain fixed before attribution

1 Diagnose

LOO boundary
 $\emptyset, D, D_{-i}$ queries

2 Enumerate

frontier F
verify minimality

3 Audit

unseen prompts
find a witness

4 Grade

causes + coverage
witness / abstain

Budget invariant: enumerate first; audit with residual physical calls

Exact-prompt cache H

one charged call per prompt class

Auditable evidence ledger

prompt ID | cost | causes | witness